CM3121 – Advanced Organic Chemistry

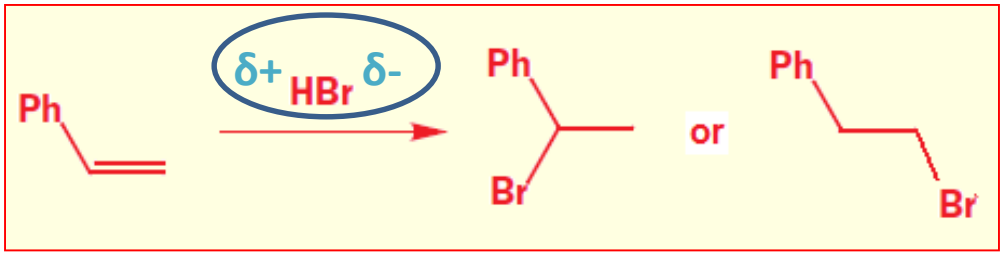
Dr. Zhu Ye

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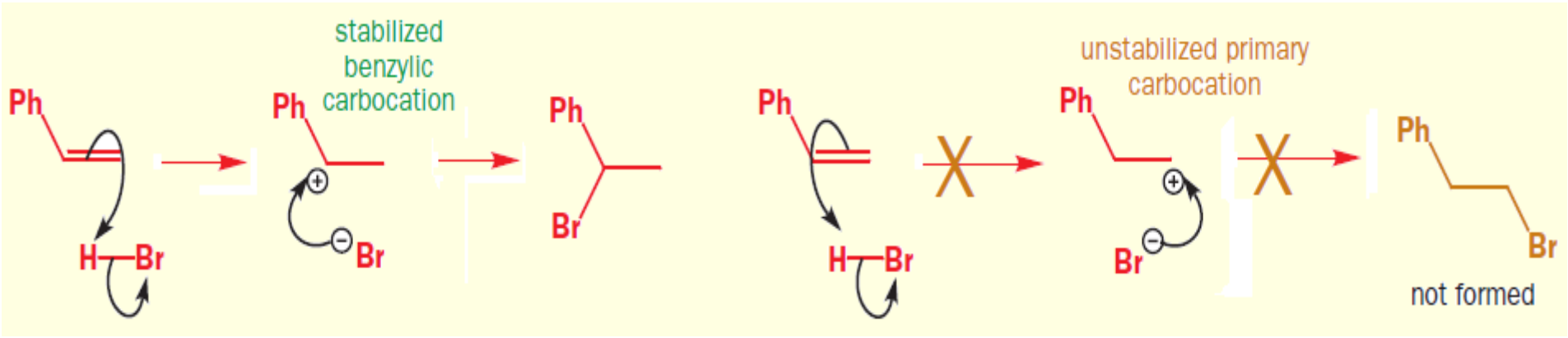
Electrophilic Addition to Alkenes (part 2)

Carbocation as the reaction intermediate and Markovnikov's rule



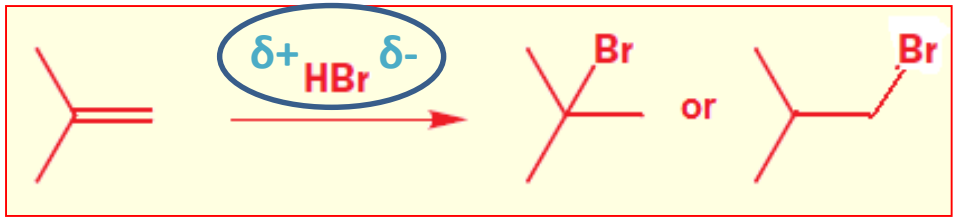
The regioselectivity of the overall electrophilic addition (H vs Br)

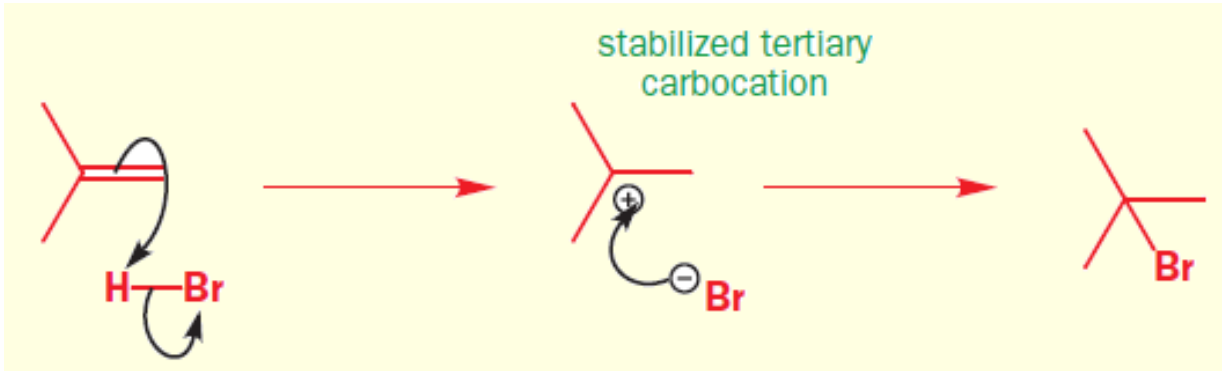
Regioselectivity: bromine atom ends up on the **more substituted** carbon



Atom labeled with "δ-" ends up on the more substituted carbon;
Atom labeled with "δ+" ends up on the less substituted carbon

Carbocation as the reaction intermediate and Markovnikov's rule





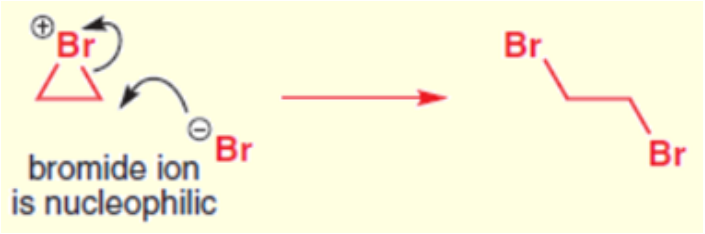
The regioselectivity of the overall electrophilic addition (H vs Br)

Markovnikov's rule: The hydrogen ends up attached to the carbon of the double bond that had more hydrogens to start with.

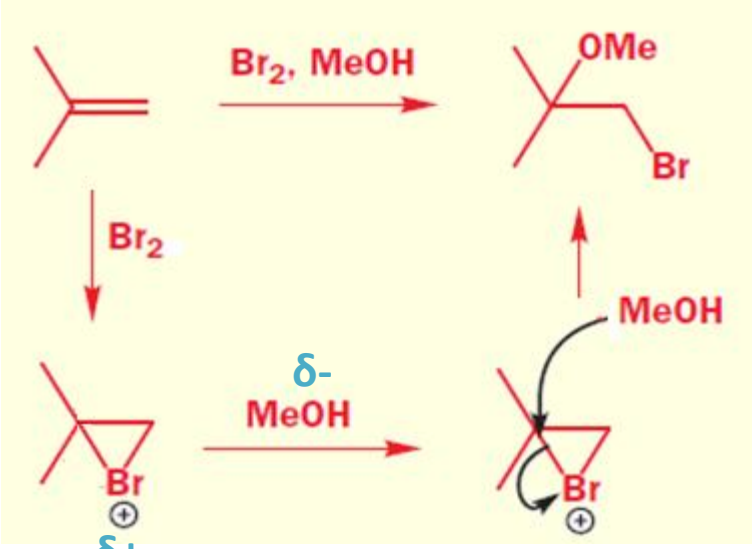
Atom labeled with "δ-" ends up on the more substituted carbon;
Atom labeled with "δ+" ends up on the less substituted carbon

Bromonium ion as the reaction intermediate

The regioselectivity is not important in the case of symmetrical bromonium ions.



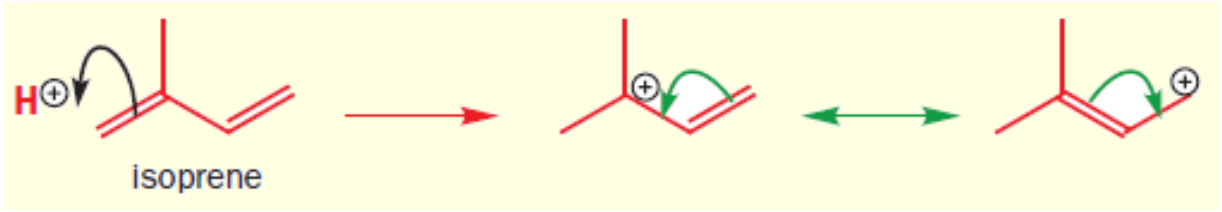
What about reactions involving **unsymmetrical Bromonium Ions?**



Observation:
An ether is formed by attack of methanol at the *more substituted* end of the bromonium ion.

Atom labeled with "δ-" ends up on the more substituted carbon;
Atom labeled with "δ+" ends up on the less substituted carbon

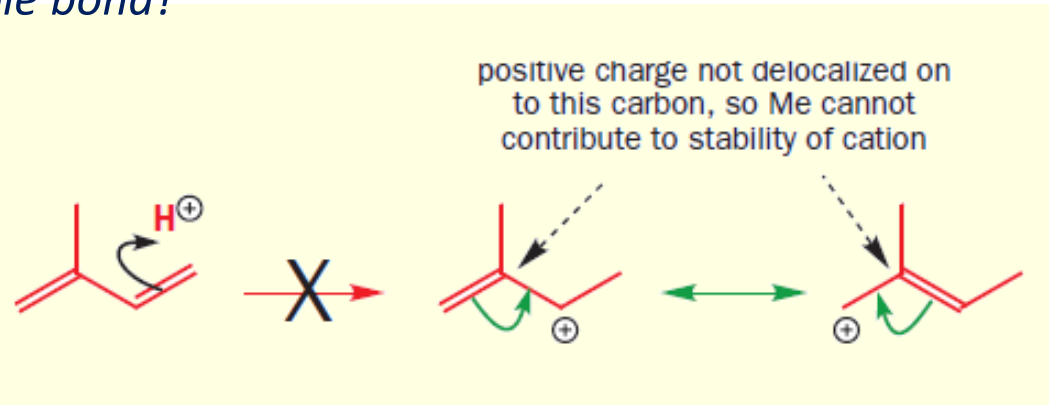
4. Electrophilic Addition to Dienes (1,4-addition)

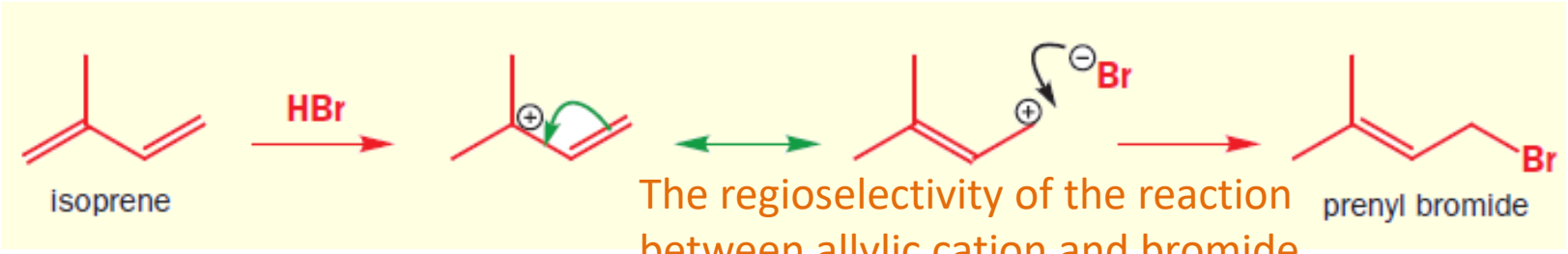


protonation gives a stable delocalized allylic cation

Why not protonate the other double bond?

Cation obtained by protonating the other double bond is also allylic, but it cannot benefit from the additional stabilization from the methyl group





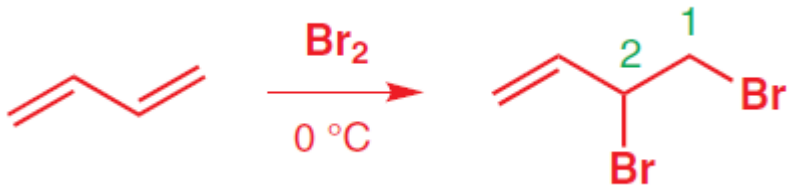
The regioselectivity of the reaction between allylic cation and bromide (the position of Br)

cation is attacked at the less hindered end

Overall, the atoms H and Br are added to the ends of the diene system



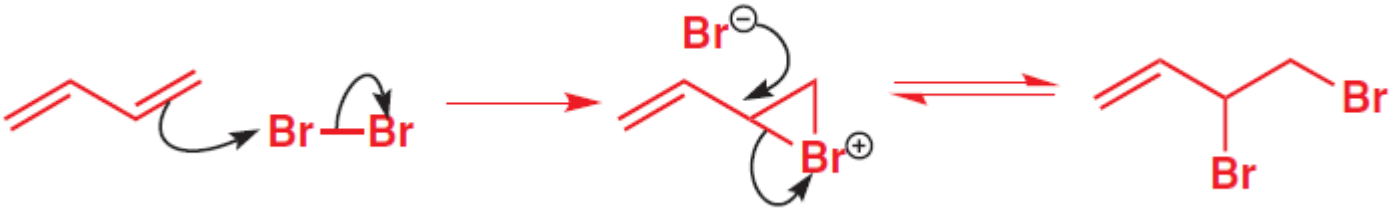
(1 equiv)



"kinetic product"



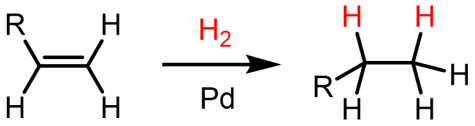
"thermodynamic product"



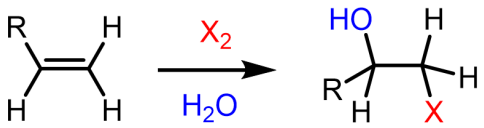


Overall, the two Br atoms are added to the ends of the diene system.

Electrophilic Addition Reactions

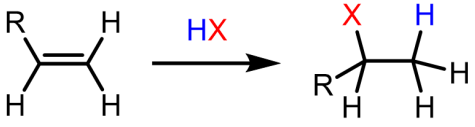


Hydrogenation



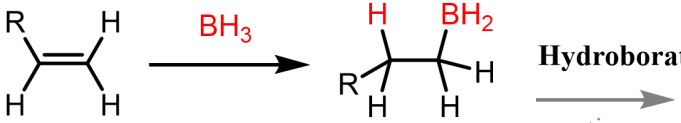
Halohydrin formation

X = Cl, Br, I



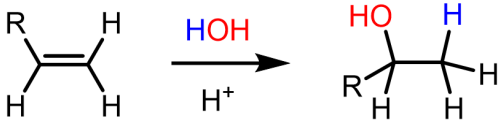
Hydrohalogenation

X = Cl, Br, I

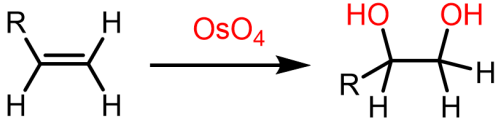


Hydroboration

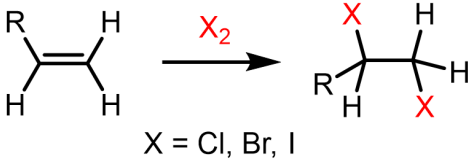
continues



Hydration

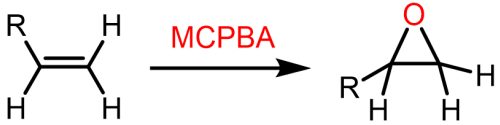


Dihydroxylation



Halogenation

X = Cl, Br, I



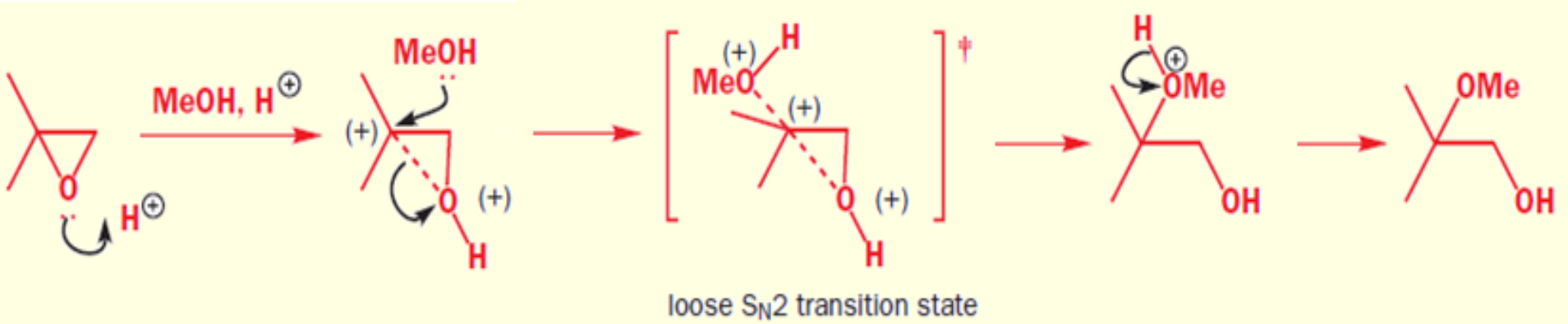
Epoxidation

6. Regioselectivity of Epoxide Opening Can Depend on the Conditions

Although epoxides, like bromonium ions, contain strained three-membered rings, they require either acid catalysis or a powerful nucleophile to react well.

acid-catalysed reaction (MeOH as the nucleophile)

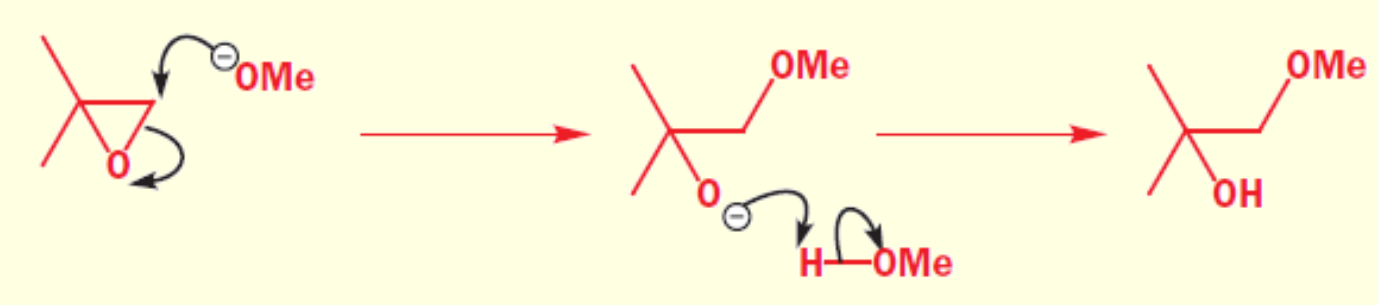
- protonation by acid produces a positively charged intermediate. The two alkyl groups make possible a build-up of charge on the carbon at the tertiary end of the protonated epoxide, and methanol attacks here, just as it does in the bromonium ion
- opening happens at the more substituted end



In this case, the regioselectivity is determined by the electronic effects (same as opening of bromonium ions)

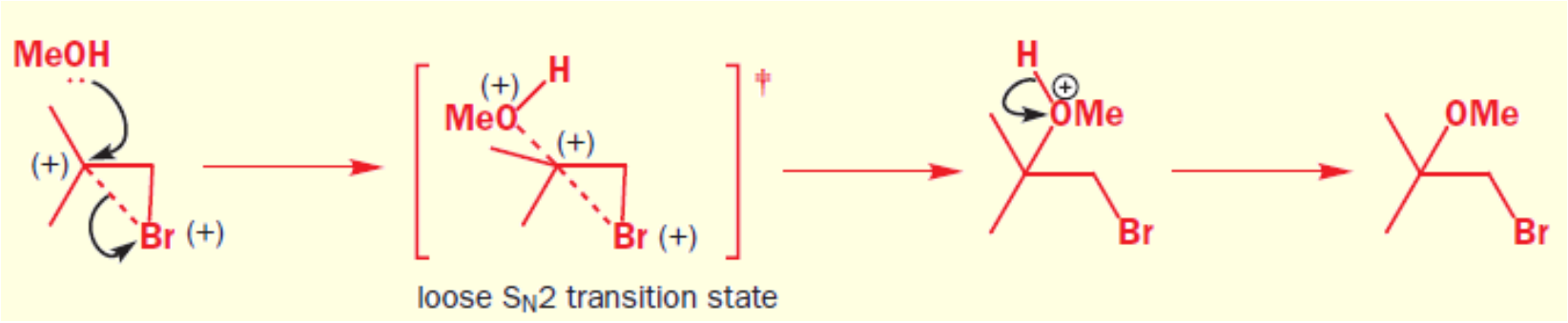
base-catalysed reaction (strong nucleophile required)

- no protonation of the epoxide, and so no build-up of positive charge;
- without protonation, the epoxide oxygen is a poor leaving group, and leaves only if pushed by a strong nucleophile: reaction becomes pure S_N2. Steric hindrance becomes the controlling factor, and methoxide attacks only the primary end of the epoxide.



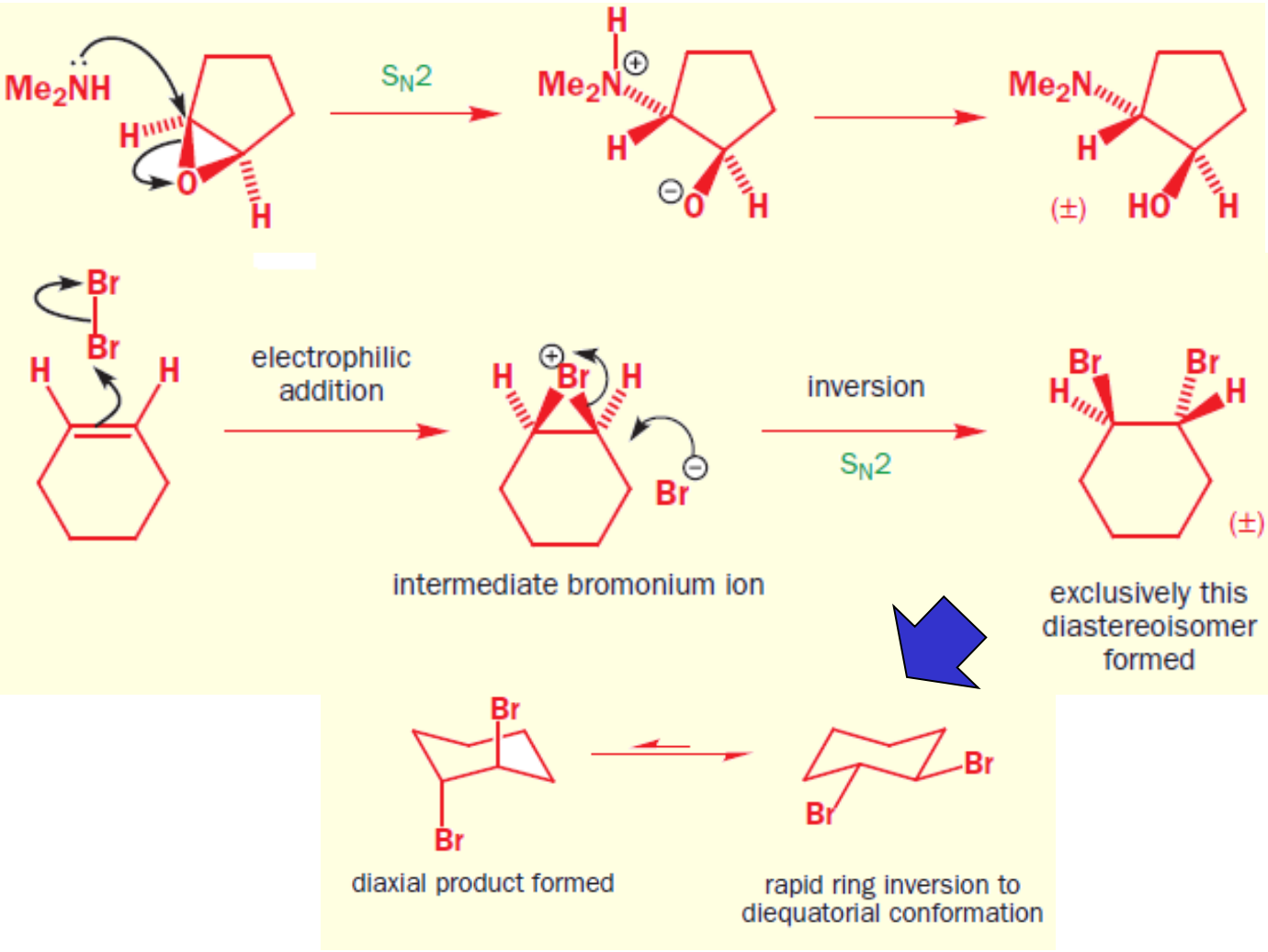
In this case, the regioselectivity is determined by the steric effects.

(compare: opening of bromonium ions)

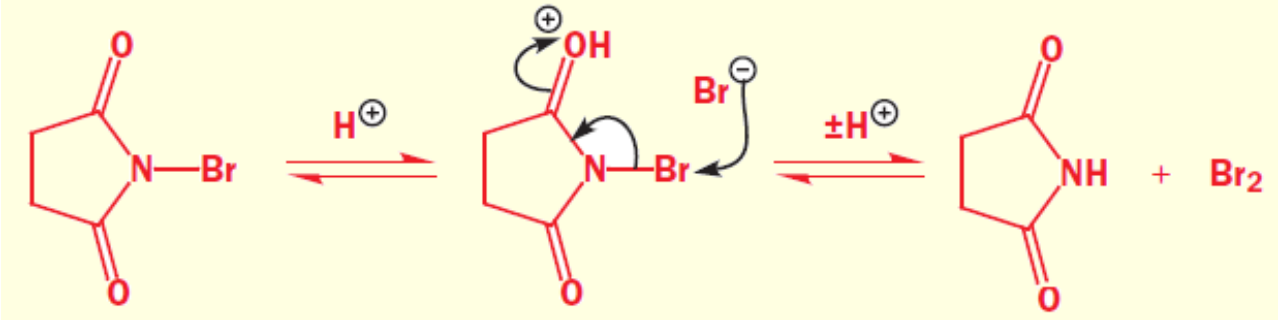


7. Electrophilic Additions to Alkenes Can be Stereoselective

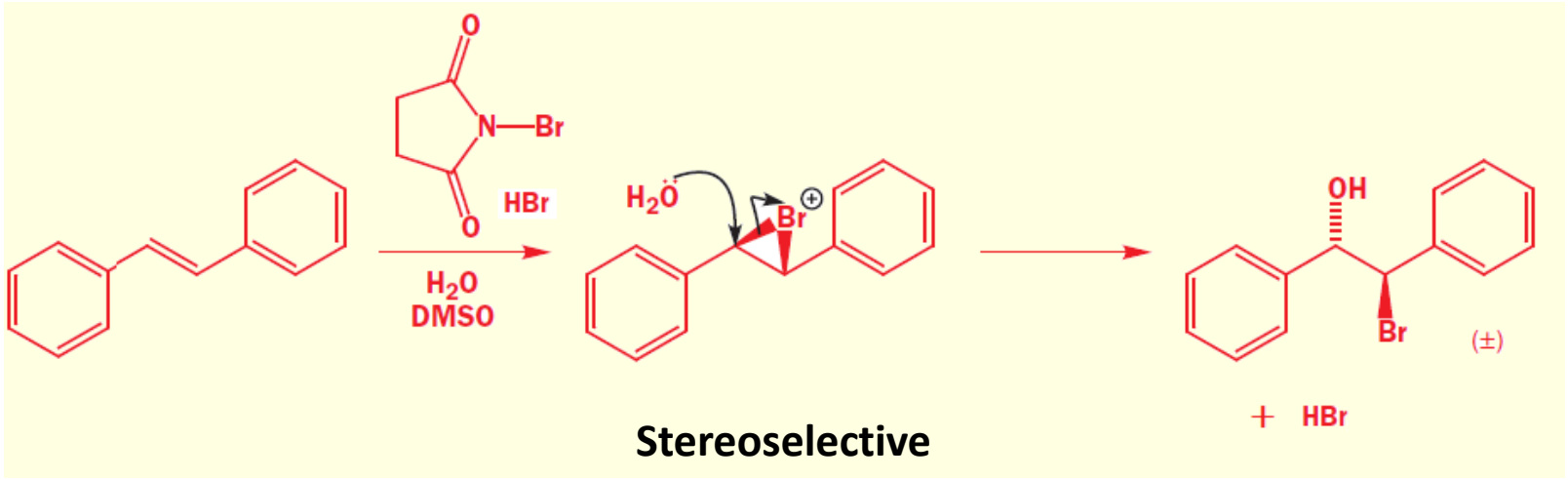
epoxide ring opening is stereospecific: it is an S_N2 reaction, and it goes with inversion



Besides bromine, *N-bromosuccinimide, or NBS* can be used as a source of Br_2 to generate the bromonium ion

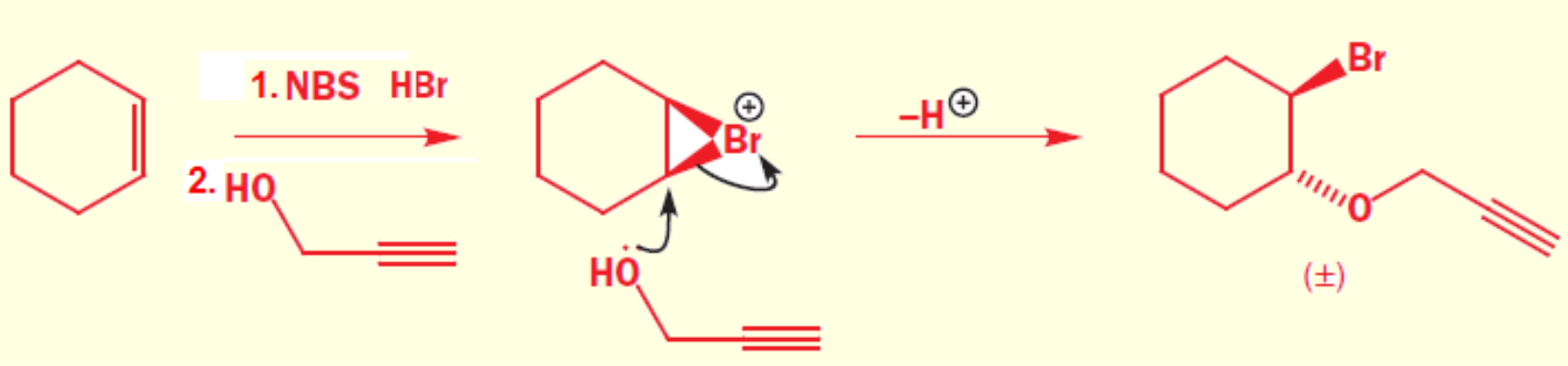


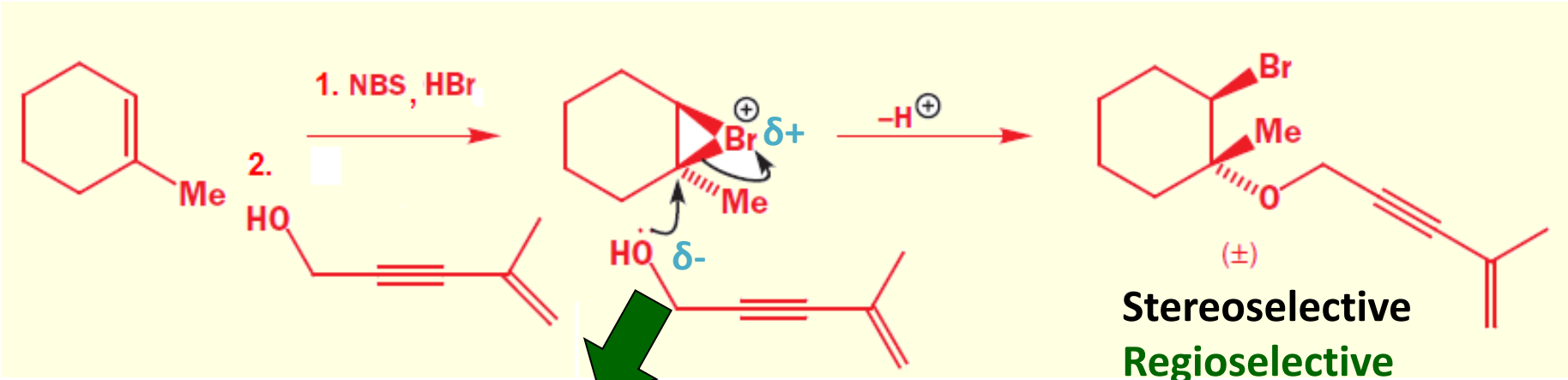
a small amount of HBr is enough to get the reaction going, and every addition reaction produces another molecules of HBr which liberates Br_2 from NBS.



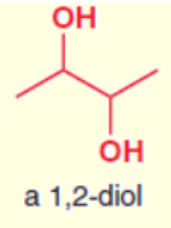
Stereoselective

Reaction between alkene, NBS, and alcohol





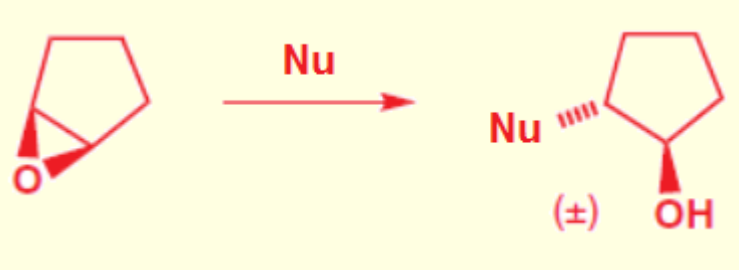
alcohol attacks the more hindered end of the bromonium ion as this provides greatest stabilization of the partial positive charge in the 'loose S_N2' transition state



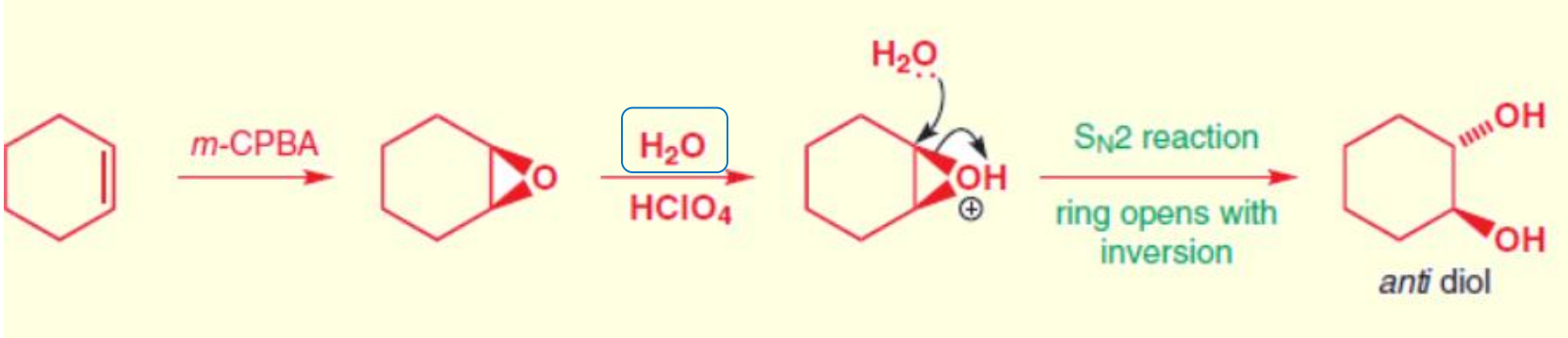
8. Making *Trans* 1,2-Diol

A good way of making 1,2-diol is to add 2 hydroxyl groups across a double bond.

Recall:



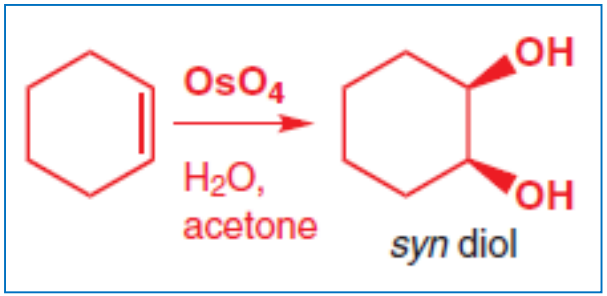
When a nucleophile opens an epoxide, it generates an alcohol



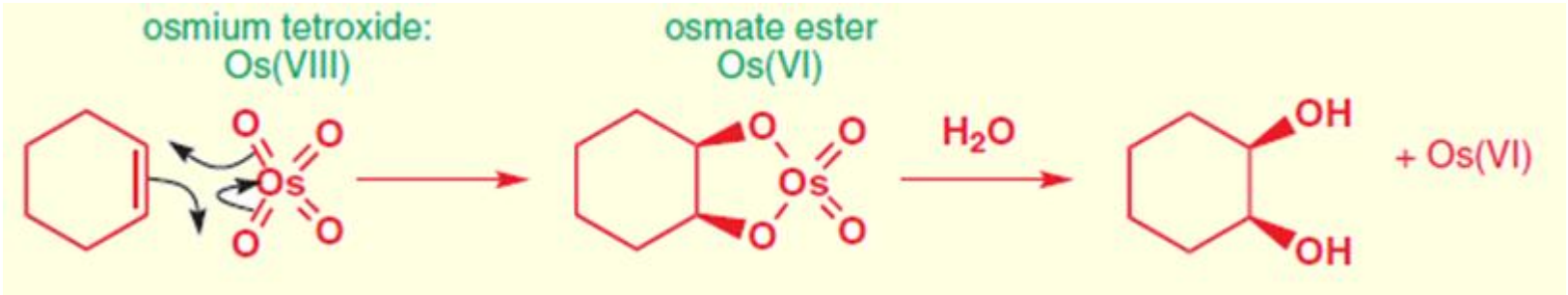
Epoxide opening goes via an S_N2 reaction with stereochemical inversion

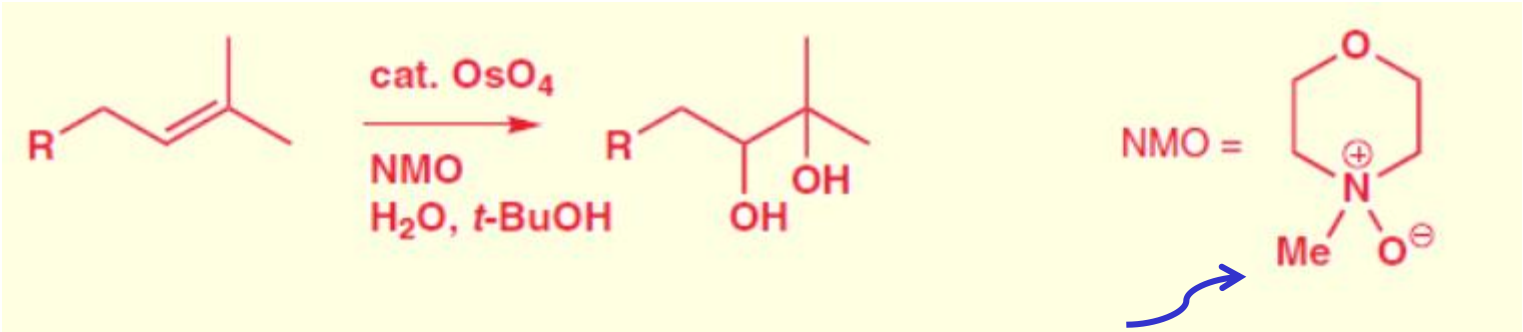
Dihydroxylation: Making *Syn* 1,2-Diol





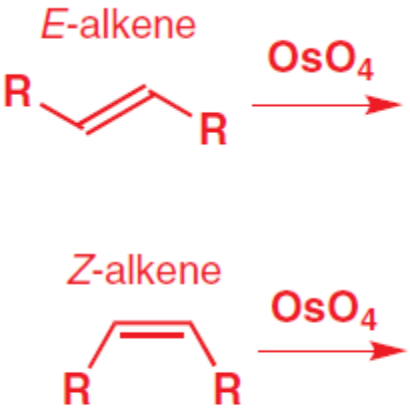
- Osmium tetroxide, OsO_4, reacts with alkenes to provide one –OH to each end of the double bond;
- Both –OH groups are delivered at the same time





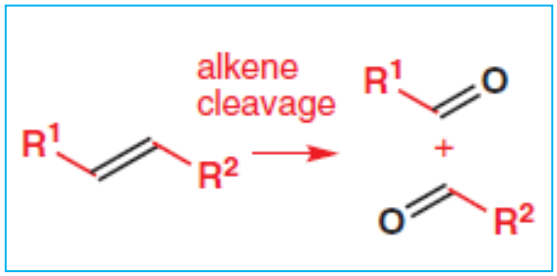
N-methylmorpholine-N-oxide (NMO) reoxidizes Os(VI) back to Os(VIII)

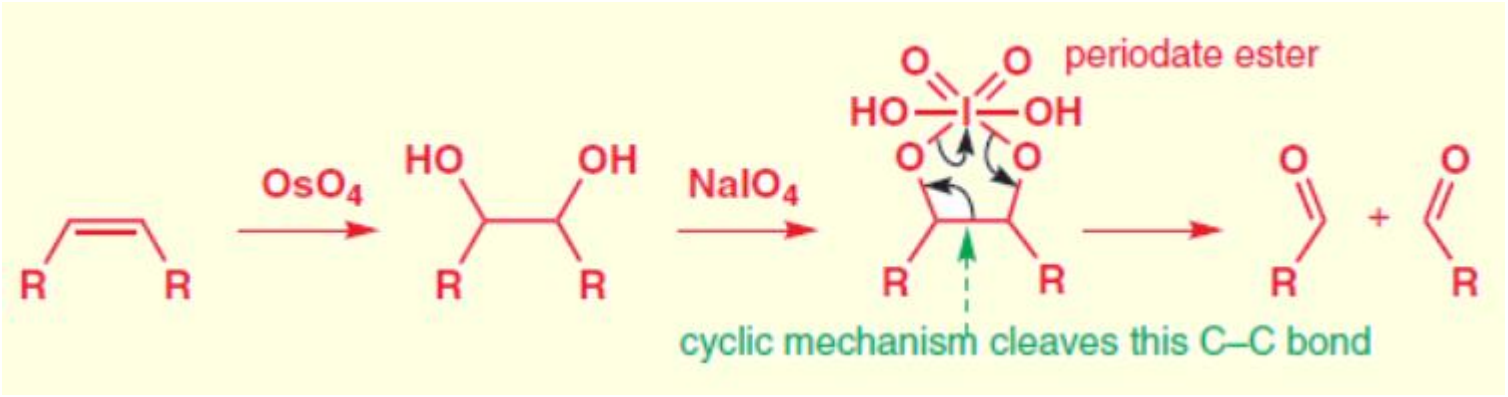
Dihydroxylation: Making *Syn* 1,2-Diol (*E*-alkene vs *Z*-alkene)



- hydroxyl group added in a *syn* fashion;
- overall product depends on the geometry of the alkene;
- reaction is stereospecific

9. Breaking a Double Bond Completely: Periodate Cleavage of 1,2-diols

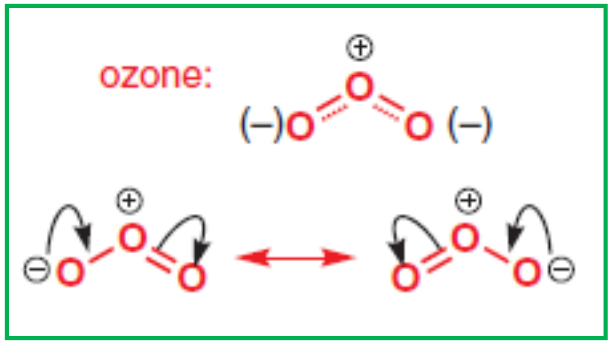




- two steps: (i) OsO_4; (ii) sodium periodate, $NaIO_4$

- reaction comprises 2 oxidation – first of the π and then the σ bond

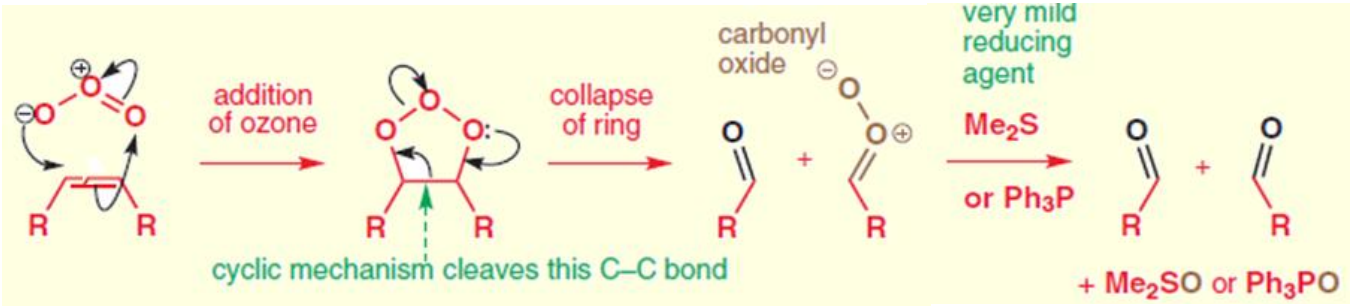
Breaking a Double Bond Completely: Ozonolysis



- ozone is unstable and is generated immediately before use from O_2 (using an ozonizer) and bubbled into the reaction mixture;

- it adds to alkenes by a cyclic mechanism;

- the 5-membered ring with 3 oxygen atoms collapses by breaking a weak O-O bond and a C-C σ bond, but gains 2 strong C=O bonds

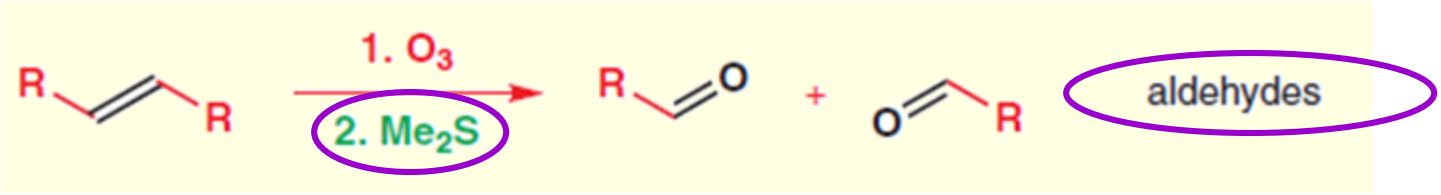


1,3-dipolar cycloaddition (discussed in future lectures)

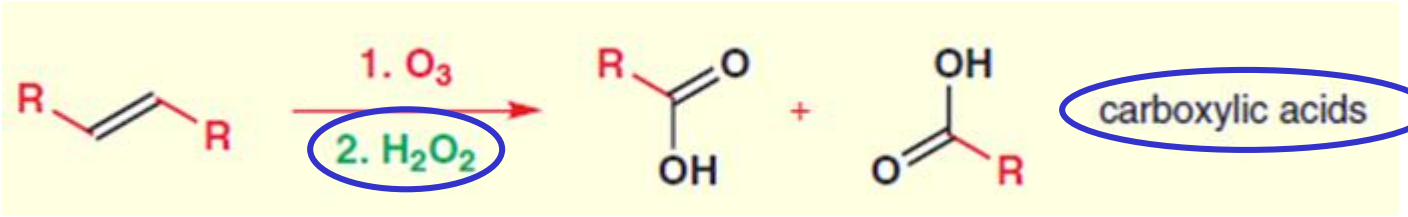
- Mild reducing agent, such as dimethyl sulfide, Me_2S, or triphenylphosphine, Ph_3P, removes the “spare” O

Breaking a Double Bond Completely: Ozonolysis

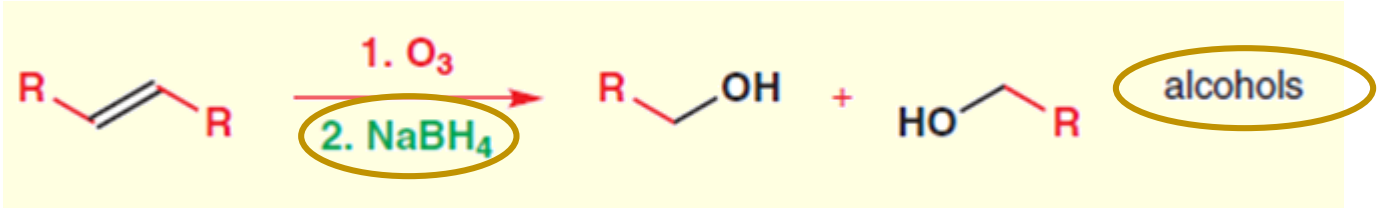
Ozonolysis can be used to generate not only aldehydes, but also other functional groups.



Mild reductant

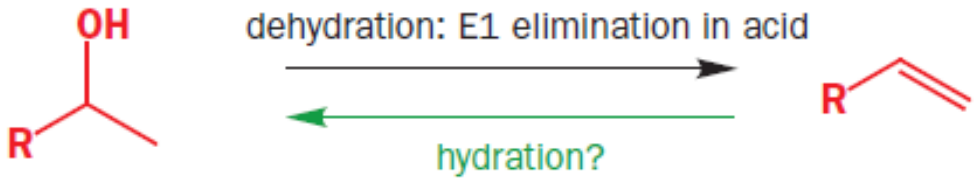


Oxidant

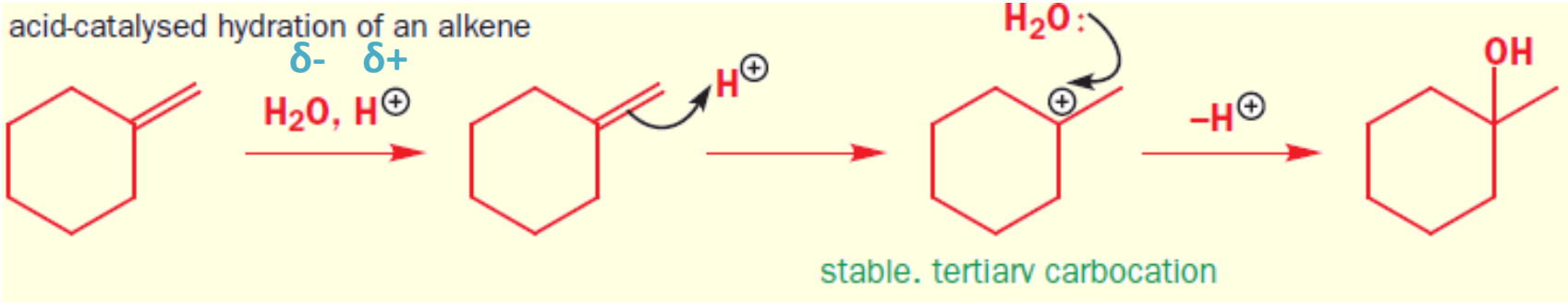


Strong reductant

10. Adding One Hydroxyl Group: How to Add Water Across a Double Bond?



The reaction works only if protonation of the alkene can give a stable, tertiary cation. The cation is then trapped by the aqueous solvent.



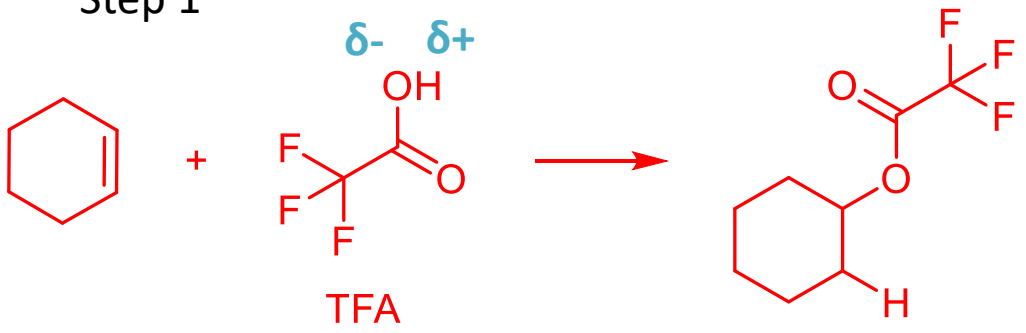
"**Markovnikov**" regioselectivity

The reaction is an equilibrium

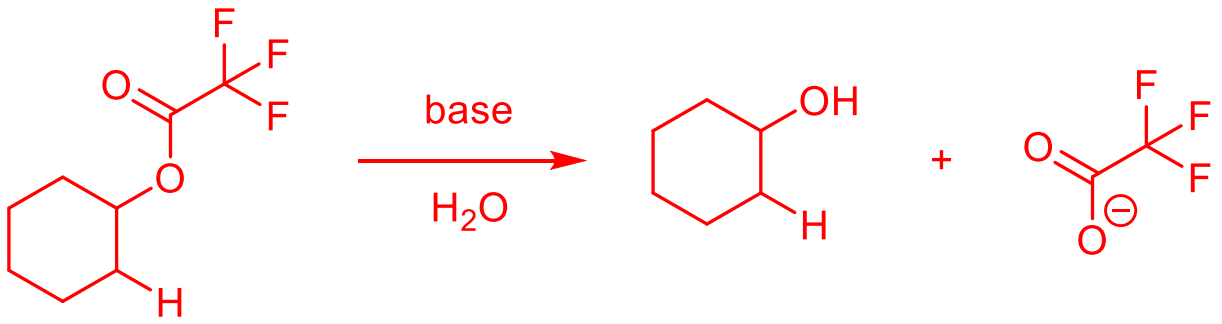


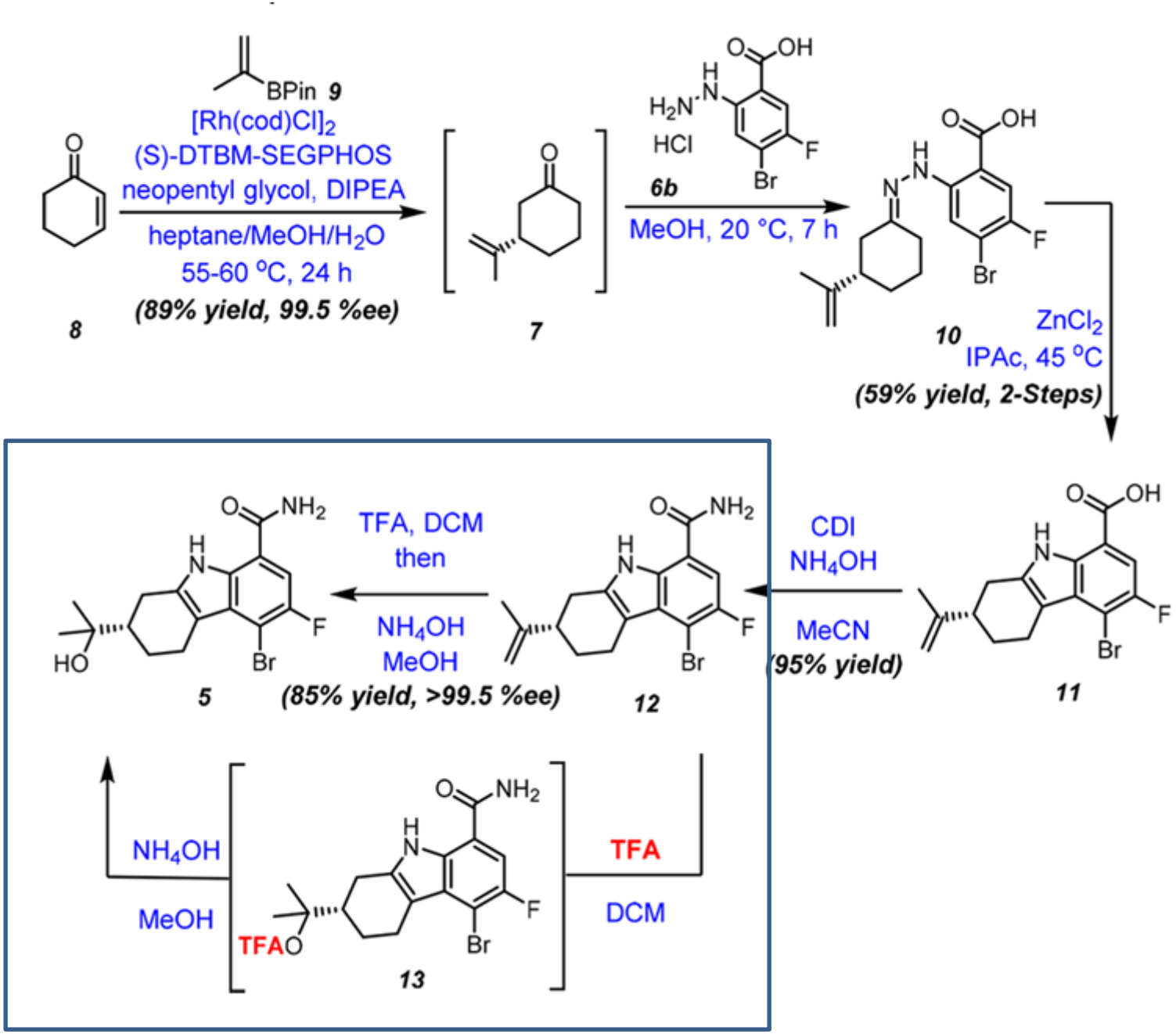
Stepwise reaction could be carried out using trifluoroacetic acid (TFA) followed by hydrolysis of the ester.

Step 1

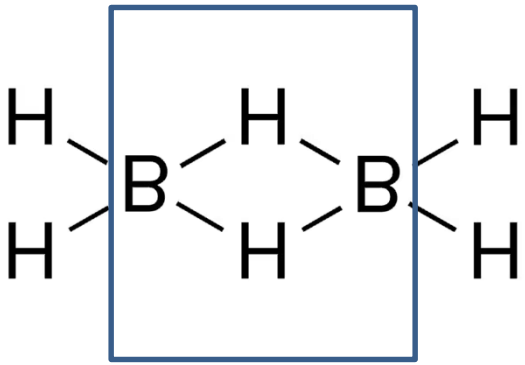


Step 2

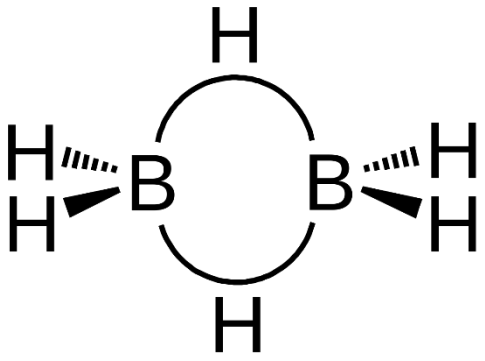



BPin 9
[Rh(cod)Cl]2
(S)-DTBM-SEGPHOS
neopentyl glycol, DIPEA
heptane/MeOH/H2O
55-60 °C, 24 h
(89% yield, 99.5 %ee)
8
7
6b
HCl
MeOH, 20 °C, 7 h
10
ZnCl2
IPAc, 45 °C
(59% yield, 2-Steps)
11
CDI
NH4OH
MeCN
(95% yield)
12
TFA, DCM
then
NH4OH
MeOH
(85% yield, >99.5 %ee)
5
NH4OH
MeOH
TFA
DCM
13

11. Hydroboration



Not ordinary bonds



The curved lines: three-center two-electron bonds. A pair of electrons bonding three atoms: two boron atoms and a hydrogen atom in the middle

HERBERT C. BROWN



Herbert C. Brown: 1979 Nobel Prize in Chemistry

Herbert C. Brown (1912-2004) was one of the leading American chemists of the 20th century, his Nobel Prize-winning work with boron compounds revolutionized synthetic organic chemistry.

His discoveries at Purdue opened entirely new avenues in both academic and industrial chemistry. Boranes are now used in the synthesis of many organic compounds, including medications such as the antidepressant Prozac and the cholesterol-lowering drug Lipitor.

Brown came to Purdue in 1947. During the late-1950s, he and his students discovered that unsaturated organic molecules can be readily converted to organoboranes through hydroboration reactions, in which boron and hydrogen add to multiple bonds.

He supervised a very large group of graduate students and post-doctoral researchers—most notably **Ei-ichi Negishi** and Akira Suzuki, 2010 Nobel laureates who studied with Brown in the 1960s. Brown retired in 1979 but continued to be an active researcher until his death in 2004.

Note that aluminum hydride is a product of this reaction. It had been previously established that diborane adds to aluminum hydride to form aluminum borohydride.[8] Clearly we had a two-stage process for adding diborane to olefins under mild conditions (14.6).

$$9RCH{=}CH_2 + Al(BH_4)_3 \xrightarrow{DG} 3(RCH_2CH_2)_3B + AlH_3$$
$$AlH_3 + \tfrac{3}{2}B_2H_6 \xrightarrow{DG} Al(BH_4)_3 \qquad 14.6$$

It then occurred to us that we might develop this into a catalytic process. All we had to do was place the olefin in the ether solvent together with a small catalytic quantity of aluminum hydride. Then, diborane could be introduced. They would form aluminum borohydride, and the latter would react with the olefin present to form the desired organoborane, regenerating the aluminum hydride (14.6). This catalytic process was tested. It worked perfectly.

We were on the point of submitting a publication reporting this new catalytic process, when we decided to run a blank experiment. We knew that diborane could not possibly react with olefins under these mild conditions. The previous work made that quite clear.[4-7] However, we thought it dramatic to include an experiment showing the remarkable change in the rate of reaction of diborane with olefins achieved by small catalytic quantities of aluminum hydride. We tried the experiment. The addition of the diborane to the olefin (14.7) proceeded just as rapidly in the absence of the aluminum hydride!

$$6RCH{=}CH_2 + B_2H_6 \longrightarrow 2(RCH_2CH_2)_3B \qquad 14.7$$

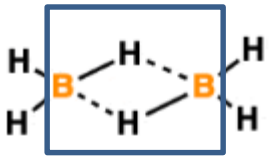
Investigation soon revealed the reaction to be rapid, quantitative, and of wide generality. A number of attempts were made to measure the rate, but without success. The reaction appeared to be complete as fast as the two reagents could be brought into contact. In its speed, generality, and quantitative nature, the reaction resembles the addition of bromine to simple olefins.

These observations were exceedingly puzzling in view of the earlier reports. Accordingly, we undertook a reexamination of the reaction of olefins with diborane in the high-vacuum apparatus. It was ob-

[8] E. Wiberg, *Angew. Chem.*, **65**, 16 (1953).

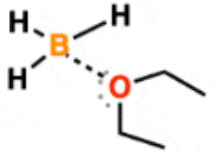
served that the reaction of diborane with pure olefins is indeed quite slow, extending over periods of many hours. However, the addition is markedly catalyzed by ethers and similar weak bases. Addition of mere traces of ethers changed the initially slow reaction to a fast one —too fast to measure by the usual manometric techniques.[9] Consequently, the inadvertent use of the ether solvents for the reactions involving sodium borohydride appears to have been largely responsible for the fast, quantitative hydroboration procedures.

Not ordinary bonds

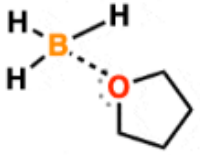


B_2H_6 (diborane)

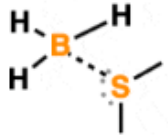
ether or week base



$BH_3 \cdot Et_2O$



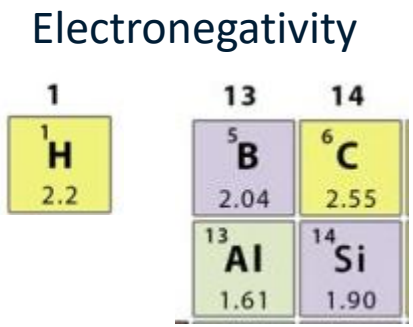
$BH_3 \cdot THF$

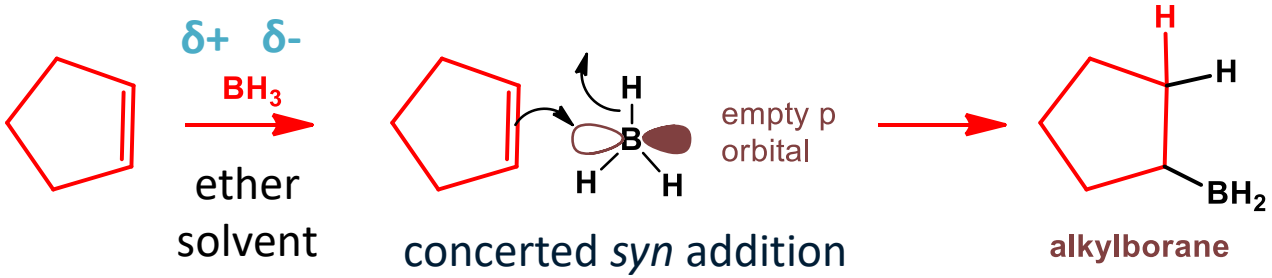


$BH_3 \cdot$ dimethylsulfide

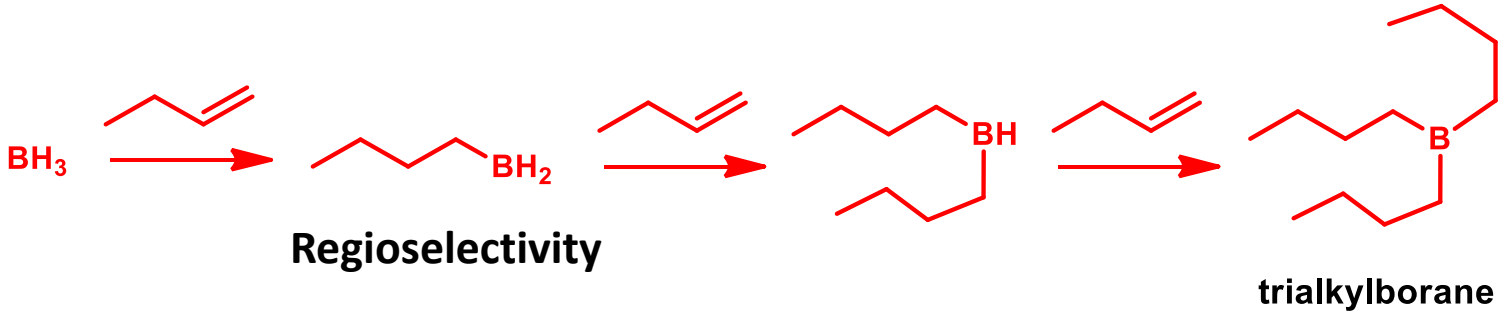
Hydroboration

Borane (BH_3 or HBR_2) adds to alkenes to make a new C-H bond and a new C-B bond





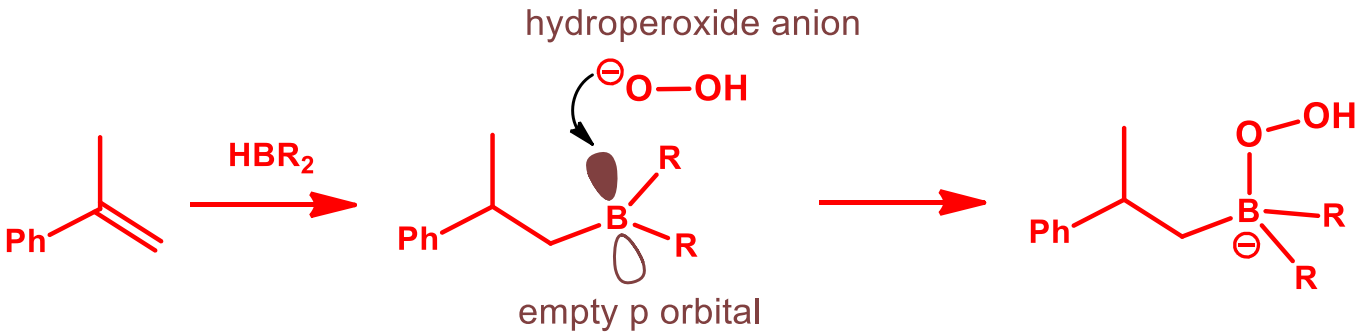
For unsymmetrical alkenes, the boron ends up on the less substituted carbon atom and the reaction can happened several times



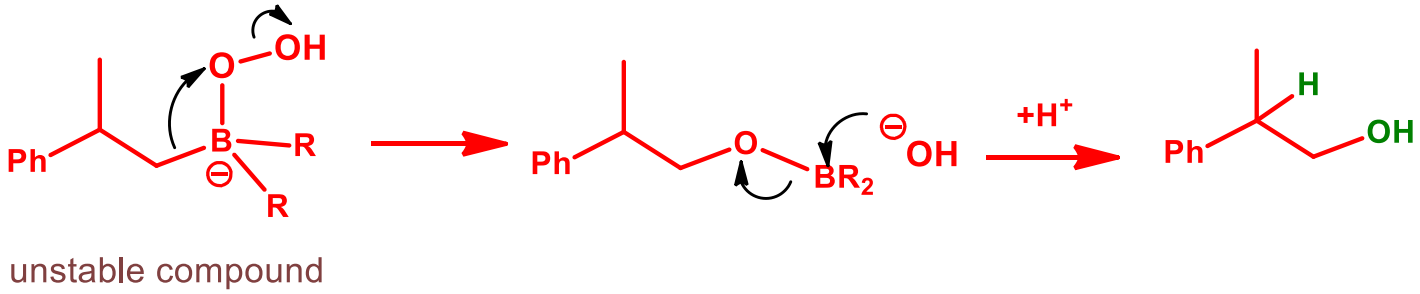
Electronegativity H > B. Seemingly, the "anti-Markovnikov" product is formed.

Oxidation of the hydroboration products

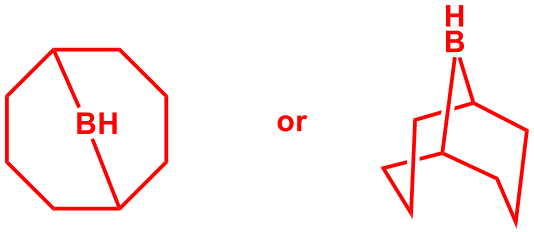
- C-B bond can be oxidized to C-O bond by a mixture of NaOH and H_2O_2
- $HO\text{-}O^-$ adds to the empty p orbital on B



- O-O bond is weak and can break, losing HO^-;
- an alkyl group on boron migrates from B to O;
- HO^- displaces B

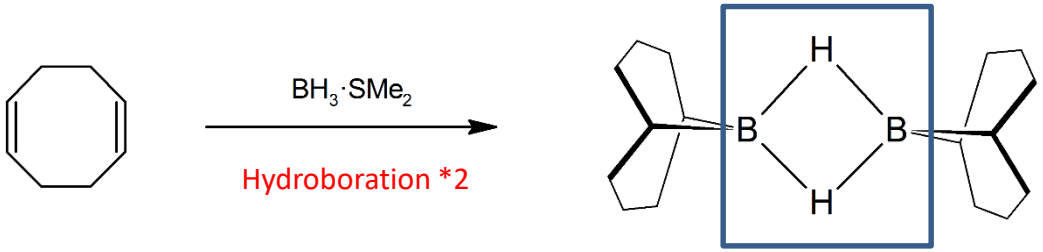


Regioselectivity of hydroboration and use of bulky borane reagent (e.g., 9-BBN)



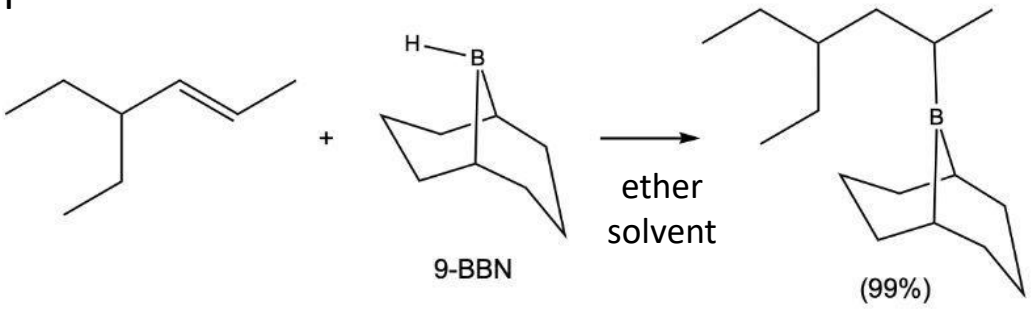
9-borabicyclononane or 9-BBN

Preparation

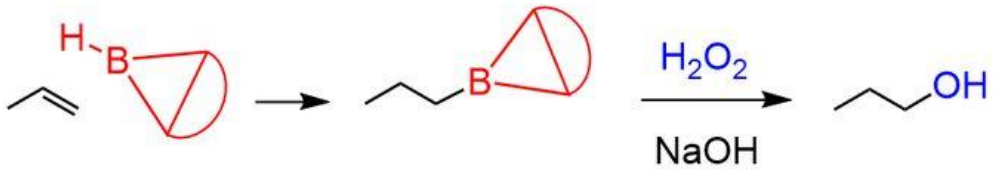


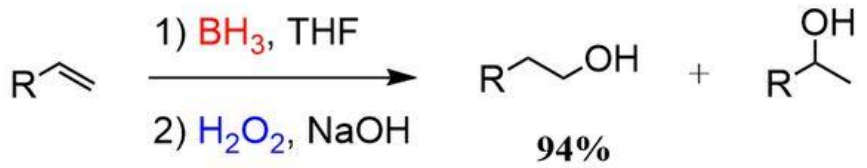
Not ordinary bonds

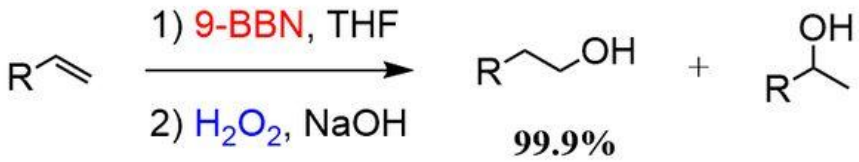
Hydroboration



Regioselectivity and bulky borane reagent 9-BBN





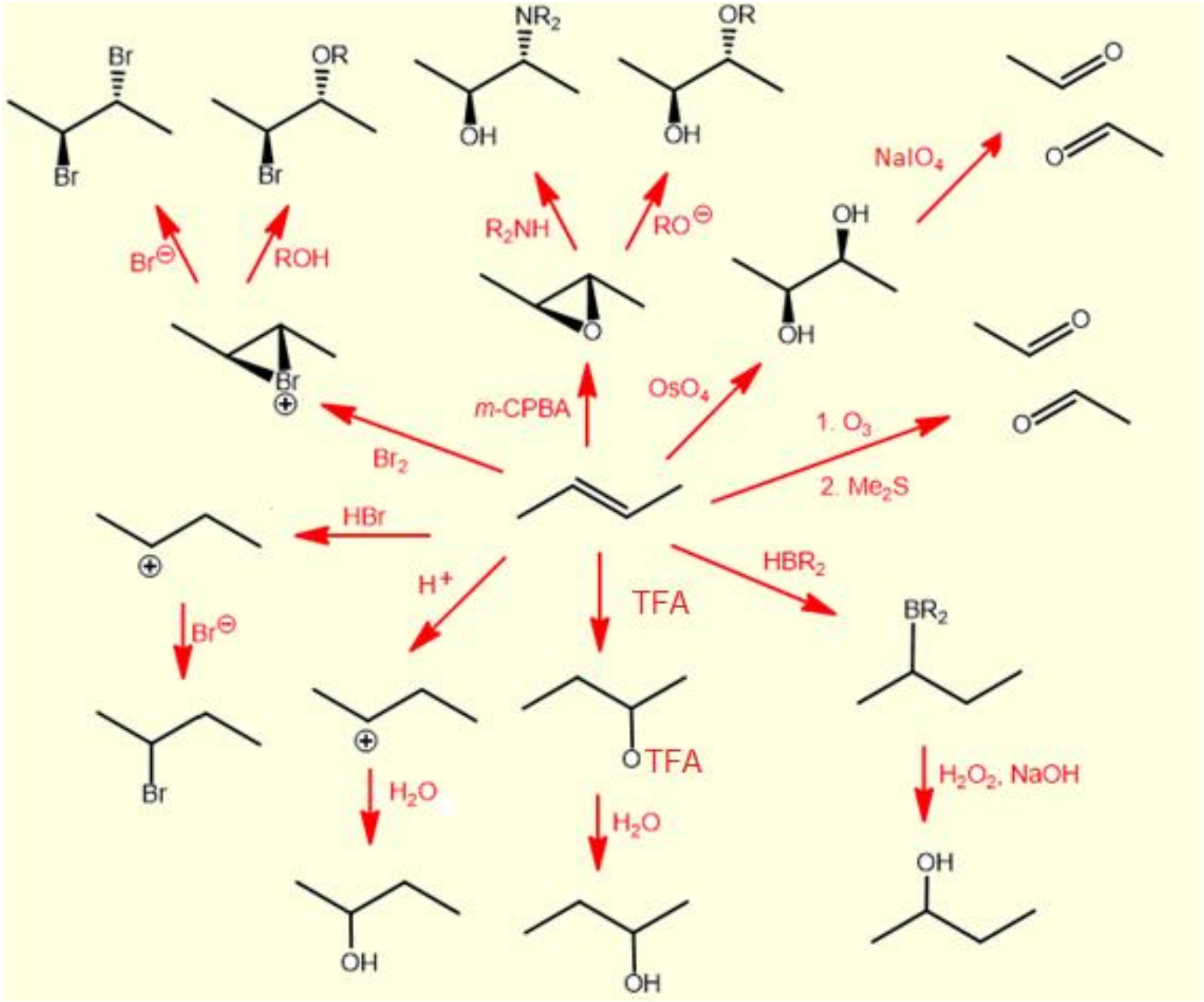


Regioselectivity:

electronic effects

electronic effects
+ steric effects

Summary


Br
Br
OR
Br
NR2
OH
OR
OH
NaIO4
O
O
Br⊖
ROH
R2NH
RO⊖
OH
OH
Br
⊕
O
m-CPBA
OsO4
1. O3
2. Me2S
O
O
Br2
HBr
H+
TFA
HBR2
⊕
Br⊖
BR2
Br
⊕
O
TFA
H2O
H2O
H2O2, NaOH
OH
OH
OH

Summary

Electrophilic addition to double bonds gives 3-membered ring intermediates with Br_2 and peroxy acids.

The 3-membered rings react with nucleophiles with (1) regioselectivity and (2) stereoselectivity.

Protonation of a double bond gives a cation, which also traps nucleophiles, and this reaction can be used to make alkyl halides (adding H-X) and alcohols (adding H-OH). “Markovnikov” product favored.

Hydroboration followed by oxidation give alcohols. Regioselectivity is seemingly “anti-Markovnikov”.

Treatment with OsO_4 then $NaIO_4$ (or O_3 then Me_2S) causes cleavage of both π and σ bonds to form carbonyl compounds.